Ningbo TengLi Testing Co., Ltd

2nd floor, Block B, Ningbo Testing and Certification Base, No. 66
Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang
Tel: 86574-8783 6802
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LM-79-19 Test Report

For

Torshare Ltd.

(Brand Name: TORSHARE)

5th Minying Road No.3 Industrial Area Shuitian Shiyan Town Bao'an District, Shenzhen, Guangdong, 518108

Model name(s): TS-LHB5-4FT300W-H-TCP[Blank;(MS)]

Report Type:	Testing and Report According to IES LM-79-2019
Type of Luminaire :	High Bay Luminaires (Commercial and Industrial)
Report Date:	2024-03-08
Prepared By:	Ningbo TengLi Testing Co., Ltd 2nd floor, Block B, Ningbo Testing and Certification Base, No. 66 Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang

Test & Report By:

Ferrum Li

Engineer: Ferrum Li

Review By:

Garman Mo

Manager: Garman Mo

Note: 1. The results contained in this report pertain only to the tested samples

2. This report does not imply product certification, approval, or endorsement by A2LA, or any agency of the Federal Government.

1.1 Product Information:		
Model Number	TS-LHB5-4FT300W-H-TCP[Blank;(MS)]	
Remark	The "[Blank;(MS)]" represents for motion sensor, it can be (MS)=motion sensor or blank=no motion sensor.	
Representative (Tested) Model	TS-LHB5-4FT300W-H-TCP(0%,4000K) TS-LHB5-4FT300W-H-TCP(100%,5000K)	
Model Difference	N/A	
SKU (if available)	N/A	
Type of Luminaire (for integral lamps, list base type and lamp type)	High Bay Luminaires (Commercial and Industrial)	
LED Manufacturer	Bridgelux Inc.	
LED Model	BXEN-27E-13H-9C1 BXEN-50E-13H-9C1	
Dimming	Continuous	
Integral Controls	Yes	
Sample Number	JDE240301-B1	
Date of Receipt	Mar.06, 2024	
Luminaire Aperture (for downlights)	--	in.
Luminaire Length	--	mm
Luminaires Width	--	mm
Number of Units (modular products)	N/A	s

1.2 Rated Values:	
Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	165W/225W/300W(Power adjustable)
Rated Initial Lamp Lumen	--
Declared CCT	4000K,5000K(Color Tunable)

1.3 Test Specifications:

Test item	1. Total Luminous Flux 2. Luminous Distribution Intensity 3. Luminous Efficacy 4. Correlated Color Temperature 5. Color Rendering Index 6. Chromaticity Coordinate 7. Electrical Parameters
Reference Standard	1. IES LM-79-2019 Optical and Electrical Measurements of Solid-State Lighting Products 2. ANSI C78.377-2008 Specifications for the Chromaticity of Solid State Lighting Products 3. CIE 13.3-1995 Method of Measuring and Specifying Colour Rendering Properties of Light Sources 4. CIE 15-2004 Technical Report Colorimetry 5. IESNA LM-16-93 Practical Guide to Colorimetry of Light Source

1.4 Test Methods

1) Photometric and Light Distribution Measurement – Goniophotometer Method: Photometric parameters were measured using the goniophotometer and software. The ambient temperature shall be maintained at 25 ℃ ±1 ℃, measured at a point not more than 1 m from the sample and at the same height as the sample. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Luminous flux, luminaire efficacy, zonal lumen were calculated from the software taken at 1 °vertical intervals and 22.5 °horizontal intervals.
2) Chromaticity Measurement – Sphere-Spectroradiometer Method: Chromaticity parameters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 ºC ±1 ºC. The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral power distribution taken at 5 nm intervals over the range of 380 to 780 nm.
3) Electrical Measurements: Electrical parameters were measured using power meters incorporated in goniophotometer or sphere-spectroradiometer system. The ambient temperature surrounding the sample was maintained at 25 ºC ±1 ºC. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Voltage, frequency, current, power, power factor and total harmonic distortion were measured by and read from the power meter.

2.1 Summary of Test Result

Criteria Item	Measured Value		Compliance	Requirement (DLC V5.1)	
Minimum Total Luminous	40877		Pass	>=10000(-10%)	
Minimum Luminous Efficacy	138.20		Pass	Standard: >= 120(-3%)	Premium: >= 135(-3%)
Minimum Power Factor	0.9377		Pass	>= 0.9(-3%)	
Maximum THD %	9.11		Pass	<= 20(+5)	
Minimum CRI	83.7		Pass	>= 70(-1)	
Minimum R9	10		Pass	>= -40(-1)	
Minimum Rg	95		Pass	>= 89(-1)	
Minimum Rf	84		Pass	>= 70(-1)	
Rcs, h1%	-13		Pass	-18%-23%(-1%)	
CCT (K)	4000K	4213	Pass	<=6500K	
	5000K	5094			
Zonal Lumen Requirement	20-50°	59.0	Pass	>=30(-10)	
Corrected UGR	26.0		Pass	Premium: <28.0	

2.2 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	Test Ambient:	25±1 °C
Test Orientation	As intended	Stabilization Time (min)	60
Model Number	TS-LHB5-4FT300W-H-TCP(0%,4000K)	Total Operating Time(min)	75

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-B1	120.0	60	2.480	295.8	0.9941	8.02
	277.0	60	1.095	284.5	0.9377	9.11

Photometric Measurement – Goniophotometer Method(Test Distance: 26.00m):

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	40877	40982
Luminous Efficacy (lm/W)	138.20	144.04
Zonal lumens in the 20-50 °	59.0	--
Beam Angle (°)	90.3	--
UGR Viewed Crosswise	26.0	--
UGR Viewed Endwise	26.0	--
Center Beam Candle Power (cd)	18552	--

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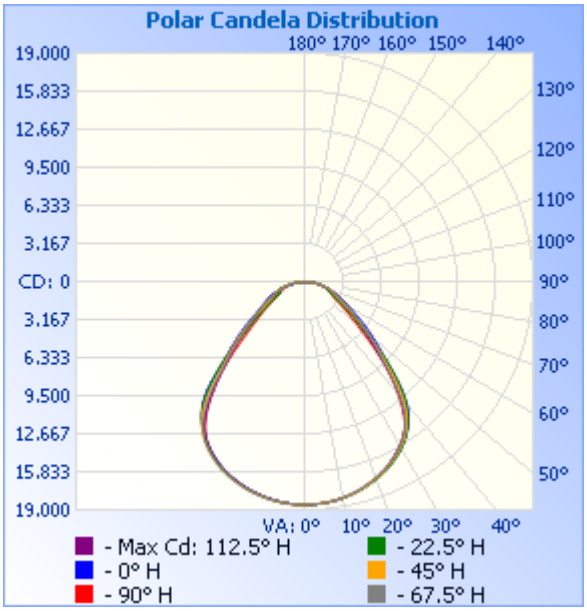
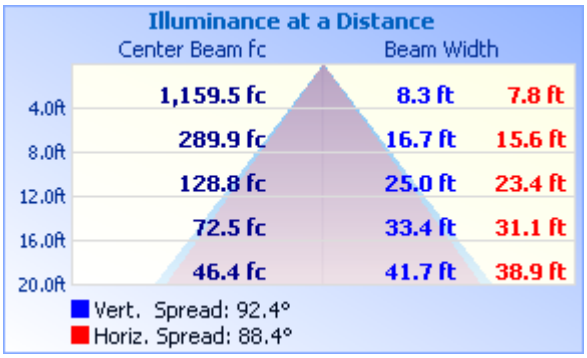
Zonal Lumen Tabulation

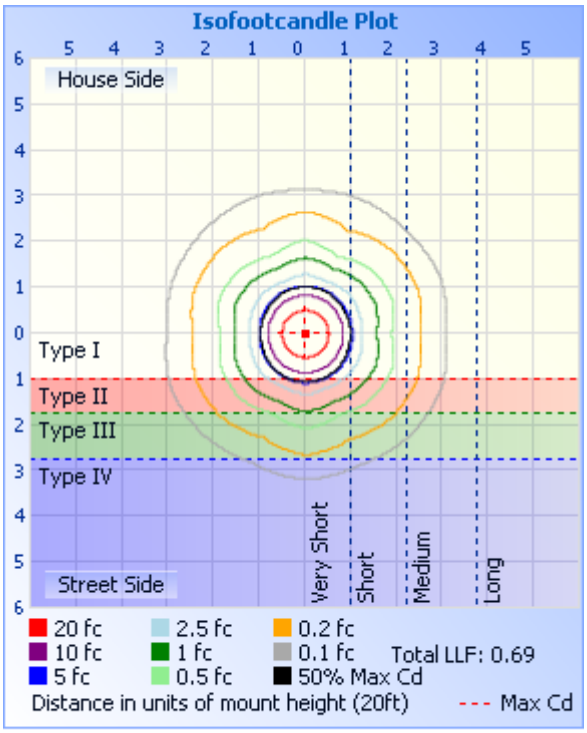
Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	14,544.0	35.6%
0-40	23,644.1	57.8%
0-60	35,399.4	86.6%
60-90	5,349.9	13.1%
70-100	2,476.2	6.1%
90-120	35.9	0.1%
0-90	40,749.3	99.7%
90-180	122.6	0.3%
0-180	40,871.8	100%

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	%Total
0-10	1,757.1	4.3%	90-100	8.0	0%
10-20	5,057.6	12.4%	100-110	11.5	0%
20-30	7,729.3	18.9%	110-120	16.4	0%
30-40	9,100.1	22.3%	120-130	19.4	0%
40-50	7,288.2	17.8%	130-140	20.8	0.1%
50-60	4,467.1	10.9%	140-150	18.9	0%
60-70	2,881.7	7.1%	150-160	14.7	0%
70-80	1,880.3	4.6%	160-170	9.2	0%
80-90	587.9	1.4%	170-180	3.7	0%

Photometric Data







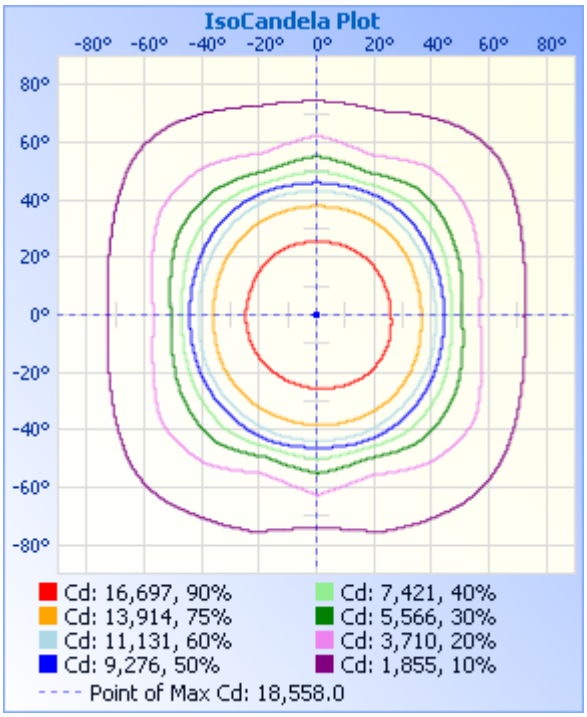


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Table--1 UNIT: X10cd

γ (DEG) \ C (DEG)	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5
0	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855	1855
5	1847	1848	1849	1850	1849	1848	1848	1846	1847	1849	1846	1845	1847	1848	1849	1850
10	1825	1826	1828	1829	1827	1827	1827	1824	1827	1827	1823	1821	1823	1827	1829	1829
15	1790	1791	1794	1793	1793	1793	1792	1787	1790	1790	1786	1783	1788	1794	1797	1798
20	1743	1742	1746	1746	1745	1748	1744	1735	1738	1737	1732	1734	1739	1747	1751	1752
25	1679	1679	1682	1685	1684	1686	1678	1669	1671	1669	1663	1666	1674	1683	1689	1692
30	1589	1589	1600	1604	1603	1604	1595	1581	1582	1580	1575	1580	1589	1598	1606	1607
35	1445	1451	1479	1492	1494	1490	1478	1455	1452	1457	1457	1462	1478	1482	1488	1475
40	1186	1183	1249	1302	1318	1309	1277	1217	1203	1223	1253	1291	1314	1307	1267	1223
45	850	881	928	985	1007	1015	961	916	897	894	912	986	980	995	932	898
50	563	659	708	655	761	686	719	712	596	661	703	618	736	629	709	672
55	405	484	525	458	577	477	526	538	421	499	515	425	546	436	512	510
60	293	380	397	319	432	339	400	419	308	406	386	294	407	307	382	411
65	235	296	298	237	334	257	310	320	248	324	311	222	335	233	306	321
70	200	240	232	198	251	211	244	249	212	253	241	212	259	221	245	241
75	157	185	177	172	185	181	189	191	167	184	184	201	158	202	184	171
80	94.9	122	123	117	130	127	131	128	105	119	131	115	91.8	118	124	118
85	39.5	50.1	60.8	56.1	72.6	61.2	63.0	53.3	43.9	48.8	60.6	52.7	63.3	51.2	56.4	50.1
90	0.44	0.49	0.59	0.89	1.15	3.91	0.88	0.52	0.32	0.38	0.44	0.50	0.60	1.19	2.50	1.51
95	0.47	0.53	0.60	1.01	1.34	0.94	0.71	0.53	0.33	0.39	0.46	0.63	0.79	0.60	0.45	0.54
100	0.51	0.62	0.94	1.20	1.77	1.15	0.89	0.59	0.38	0.44	0.61	0.80	1.15	0.77	0.62	0.49
105	0.72	0.90	1.28	1.37	1.93	1.31	1.25	0.86	0.51	0.62	0.83	1.16	1.73	1.17	0.82	0.69
110	0.99	1.23	1.59	1.93	1.84	1.82	1.55	1.19	0.66	0.79	1.11	1.64	2.50	1.79	1.07	0.88
115	1.42	1.58	2.08	1.74	1.55	1.72	2.10	1.50	0.86	1.05	1.51	1.98	2.71	2.24	1.46	1.13
120	1.90	2.04	2.74	1.56	0.73	1.73	2.65	1.92	1.07	1.39	2.02	2.12	2.52	2.34	1.89	1.39
125	2.41	2.59	3.07	3.02	0.70	2.41	2.77	2.37	1.42	1.83	2.22	2.00	1.89	2.02	2.05	1.76
130	2.95	3.08	3.19	3.21	0.80	2.48	2.79	2.77	1.96	2.26	2.55	2.65	1.64	2.87	2.49	2.16
135	3.23	3.37	3.16	3.24	0.90	2.52	2.57	2.94	2.30	2.45	3.05	3.49	1.26	3.59	2.94	2.31
140	3.30	3.41	2.86	3.27	0.99	2.51	2.69	2.98	2.55	2.74	3.23	3.99	1.35	3.95	3.13	2.63
145	3.38	3.44	2.95	3.42	1.19	2.58	3.26	2.98	2.76	2.89	3.33	4.23	1.65	3.74	3.48	2.84
150	3.39	3.38	3.60	3.49	1.29	2.60	3.92	2.98	2.87	2.92	3.65	4.35	1.79	3.26	4.24	3.07
155	3.35	3.21	4.20	3.41	1.42	2.57	4.08	2.61	2.80	2.85	3.68	4.36	2.04	2.57	4.47	3.58
160	3.20	2.96	4.31	3.22	1.45	2.47	3.93	2.56	2.76	2.72	3.52	4.12	2.41	1.86	4.48	3.57
165	3.44	3.53	4.81	3.60	1.52	2.78	4.12	3.30	2.49	2.44	3.22	3.58	2.80	1.84	3.99	3.47
170	3.84	4.25	5.58	3.86	1.77	3.20	4.95	3.94	2.95	2.87	3.87	5.23	3.59	1.93	3.98	4.63
175	4.00	4.74	5.70	3.47	1.79	2.77	4.95	4.28	3.65	3.56	4.41	5.68	3.25	1.88	3.16	5.00
180	3.58	4.36	5.19	2.91	1.60	2.26	4.55	4.08	3.67	3.56	4.30	5.27	3.02	1.64	2.37	4.71

2.3 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-LHB5-4FT300W-H-TCP(0%,4000K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-B1	120.0	60	2.482	296.1	0.9943	8.00
	277.0	60	1.096	284.8	0.9379	9.07

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.2358)(4π geometry):

Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	4213
Duv	-0.0011
Chromaticity (x, y)	x=0.3709 y=0.3682
Chromaticity (u’, v’)	u'=0.2222 v'=0.4963
Color Rendering Index (CRI)	85.6
R9	21
Rg	96
Rf	85
Rcs,h1%	-11

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	41005	41110
Luminous Efficacy (lm/W)	138.48	144.35

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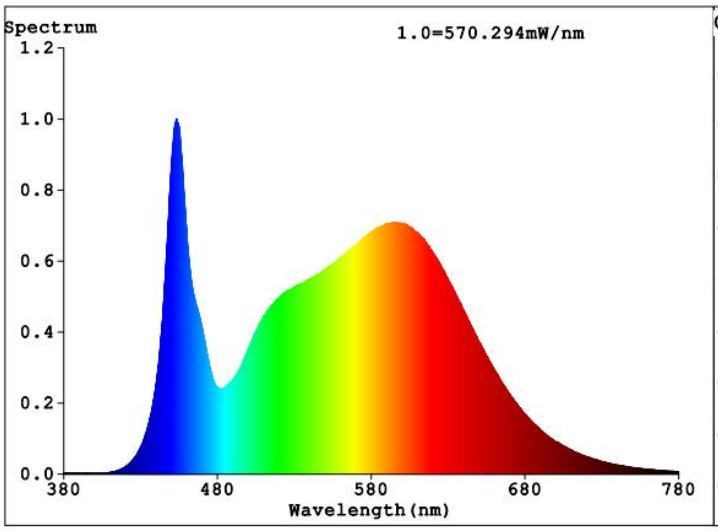
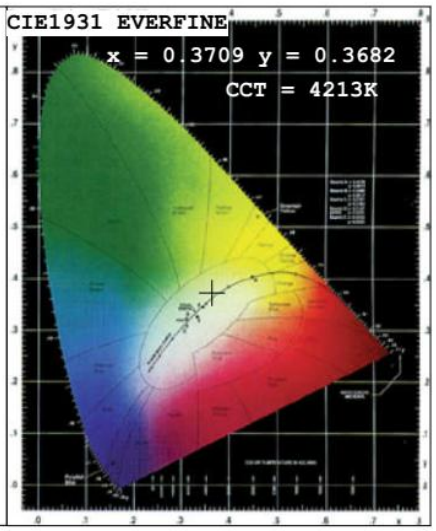
Spectral Power Distribution & Chromaticity Diagram





R1 =85	R2 =92	R3 =96	R4 =84	R5 =84	R6 =88	R7 =87	
R8 =69	R9 =21	R10=81	R11=83	R12=61	R13=87	R14=98	R15=80

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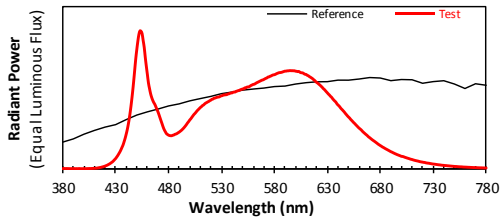
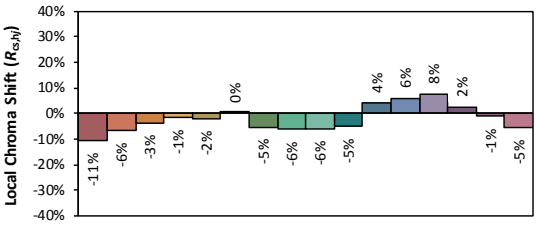
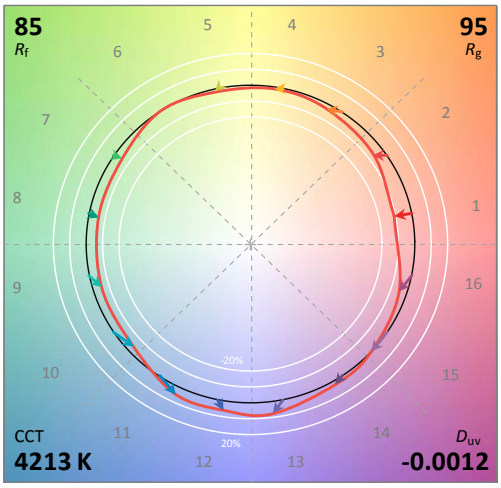
TM30

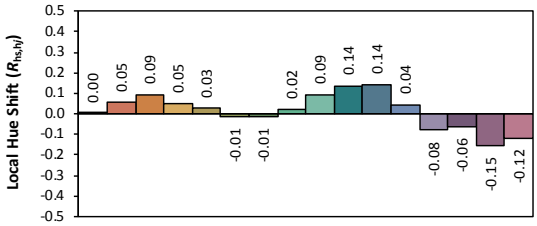
ANSI/IES TM-30-18 Color Rendition Report

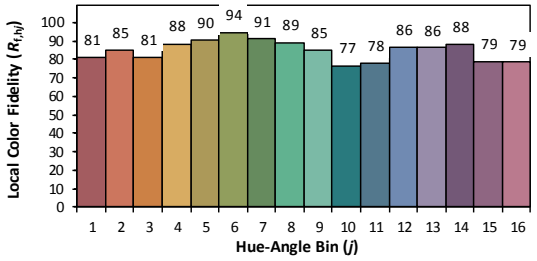
Source:	BXEN-27E-13H-9C1 BXEN-50E-13H-9C1	Manufacturer:	Torshare Ltd.
Date:	2024-03-07	Model:	TS-LHB5-4FT300W-H-TCP(0%, 4000K)

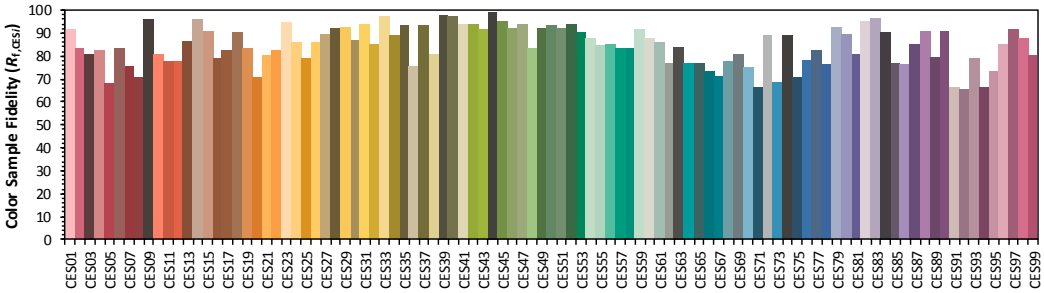












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3708
y	0.3681
u'	0.2222
v'	0.4963

CIE 13.3-1995 (CRI)

R_a	86
R_9	21

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

2.4 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-LHB5-4FT300W-H-TCP(100% ,5000K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-B1	120.0	60	2.474	295.4	0.9952	7.95
	277.0	60	1.092	284.0	0.9391	8.98

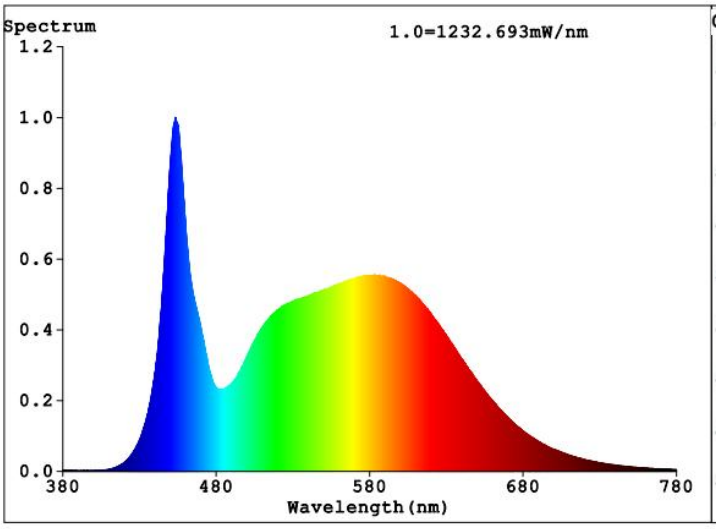
Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.2359)(4π geometry):

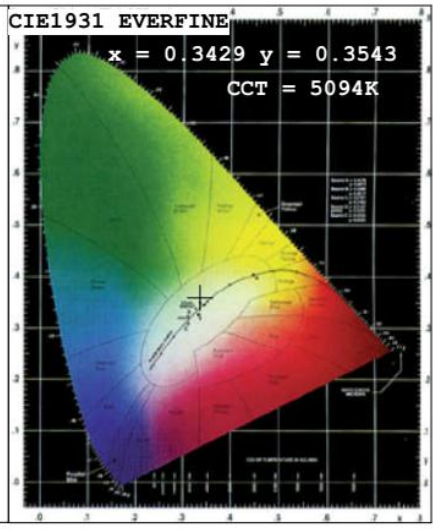
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	5094
Duv	0.0023
Chromaticity (x, y)	x=0.3429 y=0.3543
Chromaticity (u’, v’)	u'=0.2089 v'=0.4857
Color Rendering Index (CRI)	83.7
R9	10
Rg	95
Rf	84
Rcs,h1%	-13

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	41788	41895
Luminous Efficacy (lm/W)	141.46	147.52

Spectral Power Distribution & Chromaticity Diagram





R1 =82	R2 =90	R3 =94	R4 =82	R5 =82	R6 =85	R7 =87	
R8 =67	R9 =10	R10=75	R11=81	R12=60	R13=84	R14=97	R15=77

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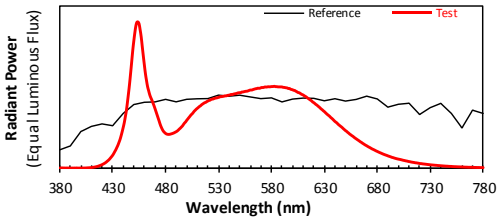
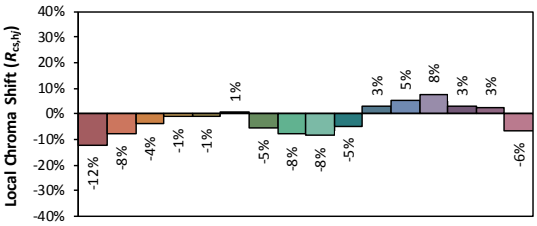
Fax: 86574-8783 5902

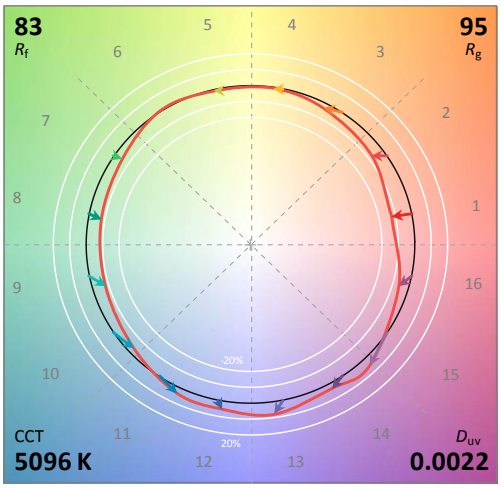
TM30

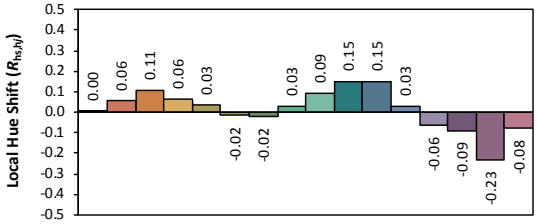
ANSI/IES TM-30-18 Color Rendition Report

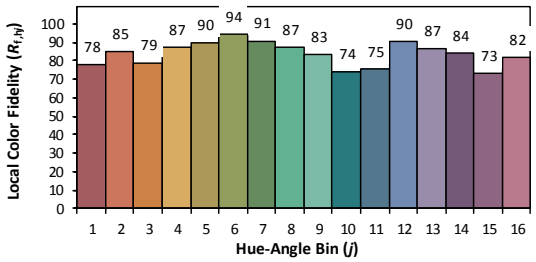
Source:	BXEN-50E-13H-9C1	Manufacturer:	Torshare Ltd.
Date:	2024-03-07	Model:	TS-LHB5-4FT300W-H-TCP(100%,5000K)

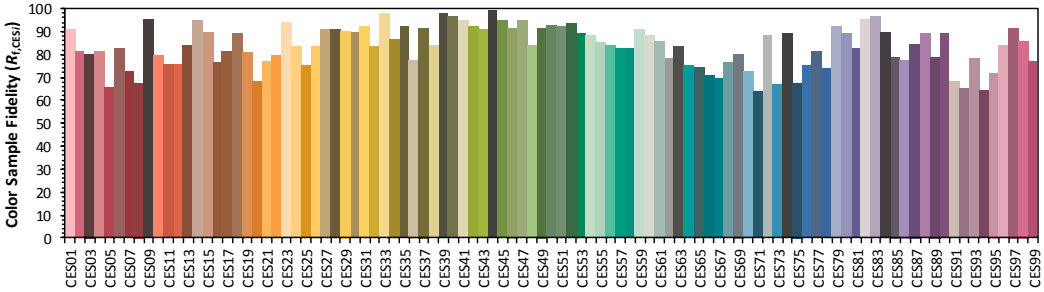












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3428
y	0.3542
u'	0.2089
v'	0.4856

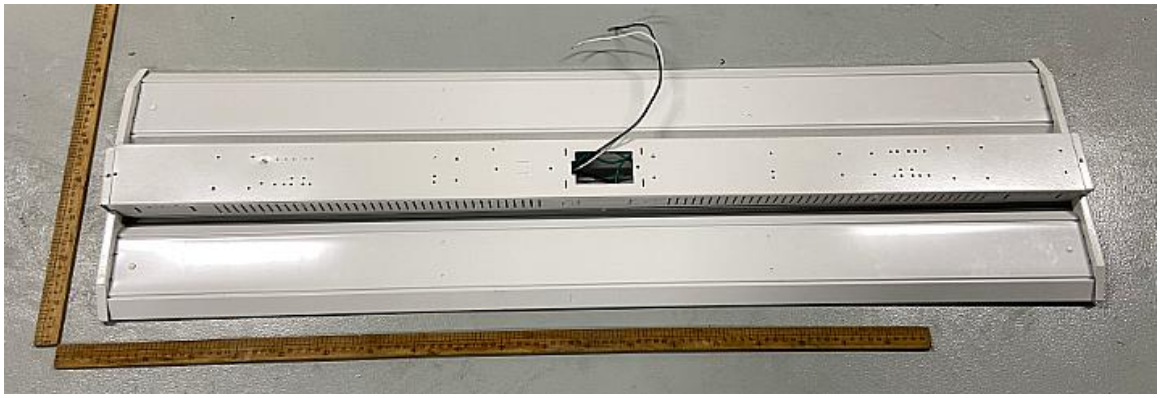
CIE 13.3-1995 (CRI)	
R_a	84
R_9	10

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

3. Test Equipment

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-702	2 meter Integrating Sphere	Verified by D204 standard lamp	
ST-R-701	Spectral analysis system HAAS-1200	Verified by D204 standard lamp	
ST-R-703	Standard Lamp D204	2023-12-26	2024-12-25
ST-R-704	Power Meter for Integrating Sphere	2023-12-26	2024-12-25
ST-R-707	Temperature Probe for Integrating Sphere	2023-12-26	2024-12-25
ST-R-714	Goniophotometer system	Verified by D908S standard lamp	
ST-R-710	Standard Lamp D908S	2023-12-26	2024-12-25
ST-R-711	Power Meter for Goniophotometer	2023-12-26	2024-12-25
ST-R-709	Hygrothermograph for Goniophotometer	2023-12-26	2024-12-25
Uncertainty(K=2): Photometric Measurement (Sphere):3.40% Chromaticity Measurement(Sphere):44.8K Photometric Measurement(Goniophotometer):3.64%			

4. Product Photo



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